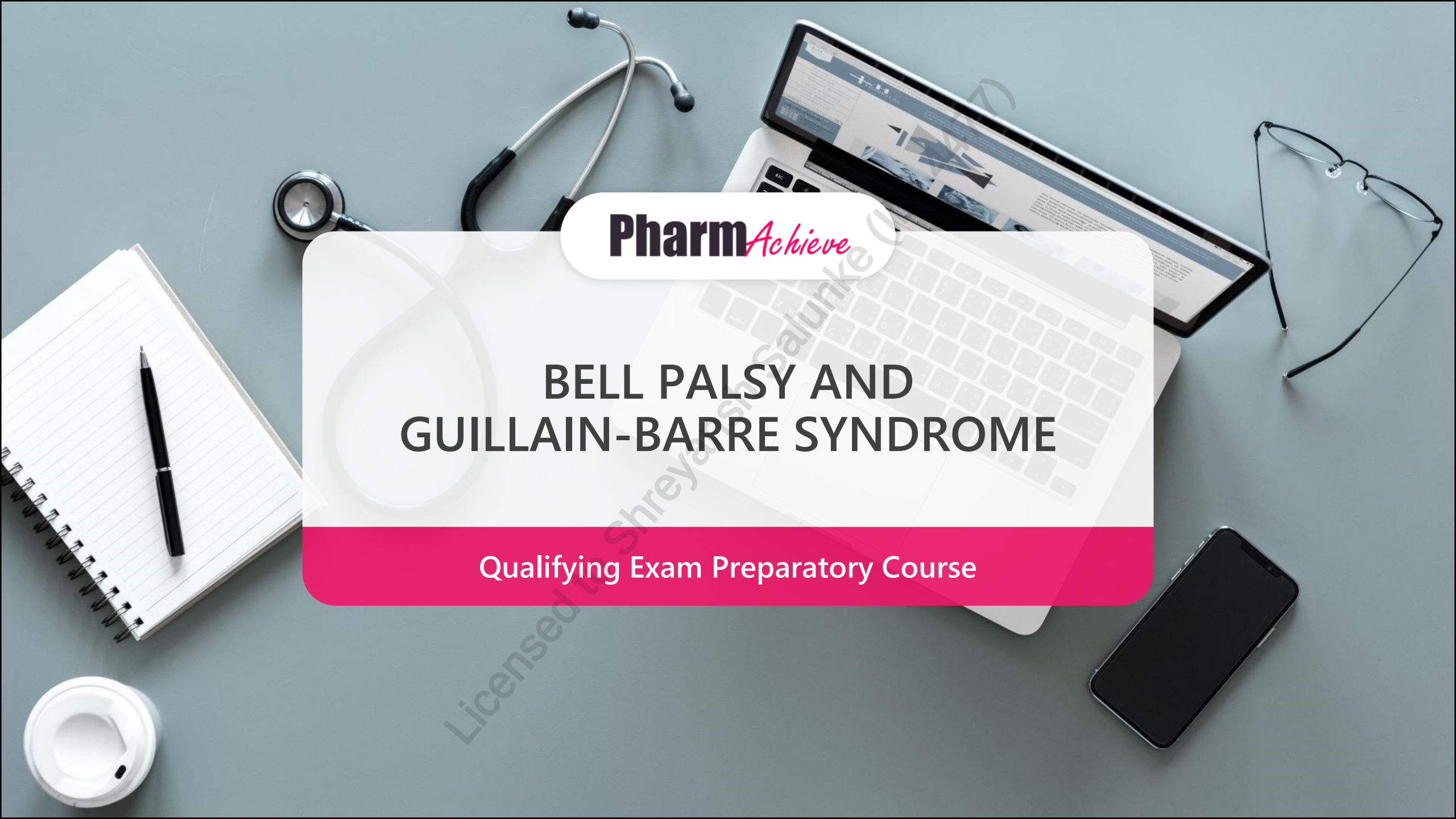


A top-down view of a grey desk with various items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill-shaped graphic is centered over the laptop.

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BELL PALSY AND GUILLAIN-BARRE SYNDROME

Qualifying Exam Preparatory Course

COVERED IN THIS PRESENTATION...

Ethical, Legal and
Professional
Responsibilities

Patient Care

Product
Distribution

Practice Setting

Health Promotion

Knowledge and
Research
Application

Communication
and Education

Intra and Inter-
Professional
Collaboration

Quality and Safety

LEGEND

BP	Bell Palsy
DVT	Deep Vein Thrombosis
GBS	Guillain-Barre Syndrome
GI	Gastrointestinal
HSV	Herpes Simplex Virus
ICU	Intensive Care Unit
LMWH	Low Molecular Weight Heparin
RBC	Red Blood Cells
WBC	White Blood Cells

Licensed to Shreyansh Salunke (ID: 4477)

BELL PALSY

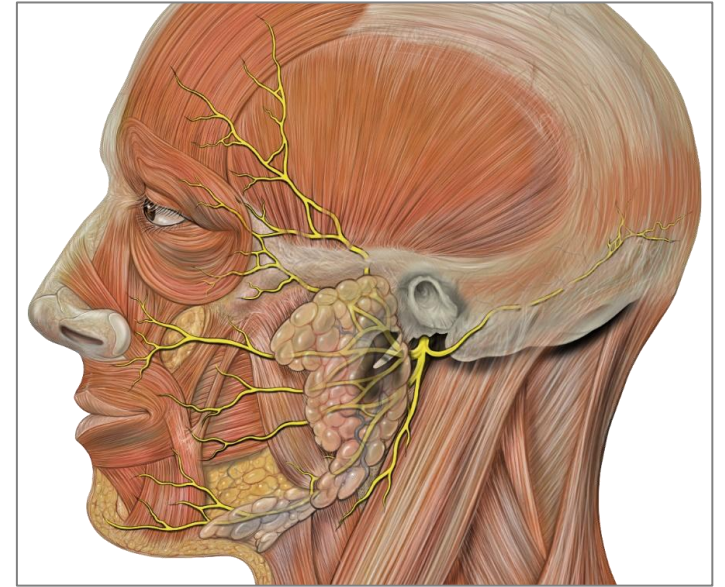
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WHAT IS BELL PALSY?

- Acute onset of unilateral facial weakness or paralysis due to infected facial nerve
- Paralysis occurs when the infected facial nerve swells due to inflammation caused by viral infection
- This facial nerve controls the facial muscles, taste, tears, saliva and a bone in the middle ear
- Rarely affects both sides of the face
- Direct cause is unclear however it is often linked to a viral infection, most commonly herpes simplex virus-1 (cold sores, genital herpes)
- Other associated viruses: adenovirus (respiratory conditions), coxsackievirus, cytomegalovirus, Epstein-Barr virus, rubella virus, herpes zoster virus (shingles and chicken pox), influenza B virus and mumps



CLINICAL PRESENTATION

- Reduced ability to taste
- Changes in tear and saliva production
- Difficulty closing the affected eye
- Difficulty making facial expressions (e.g. smiling)
- Drooping of one corner of the mouth and drooling
- Ear pain and an increased sensitivity to sound
- Headaches
- Pain around the jaw
- Weakness or paralysis on one side of the face
- Eyebrow that sags on one side

RISK FACTORS

- Patients with diabetes
- Patients with hypertension
- Patients with an upper respiratory tract infection (common cold or flu)
- Obesity
- Pregnant women (most commonly in the 3rd trimester or during the first few weeks after delivery)
- Pre-eclampsia

GOALS OF THERAPY

- ✓ Alleviate pain
- ✓ Promote complete facial recovery
- ✓ Protect the eye from damage, specifically corneal ulceration and permanent visual impairment
- ✓ Prevent irreversible facial nerve damage

NON-PHARMACOLOGICAL MEASURES

Protect the affected eye, especially when patient is unable to keep it closed

Keep eye moist by using lubricant eyedrops or an eye ointment

- Lubricant drops should be applied frequently during the day and an eye ointment should be used at night

Wear sunglasses or goggles during the daytime

Gently tape the eyelid closed during sleep

Exercise and physiotherapy may be suggested for patients with chronic persistent weakness

PHARMACOLOGICAL MEASURES: CORTICOSTEROIDS

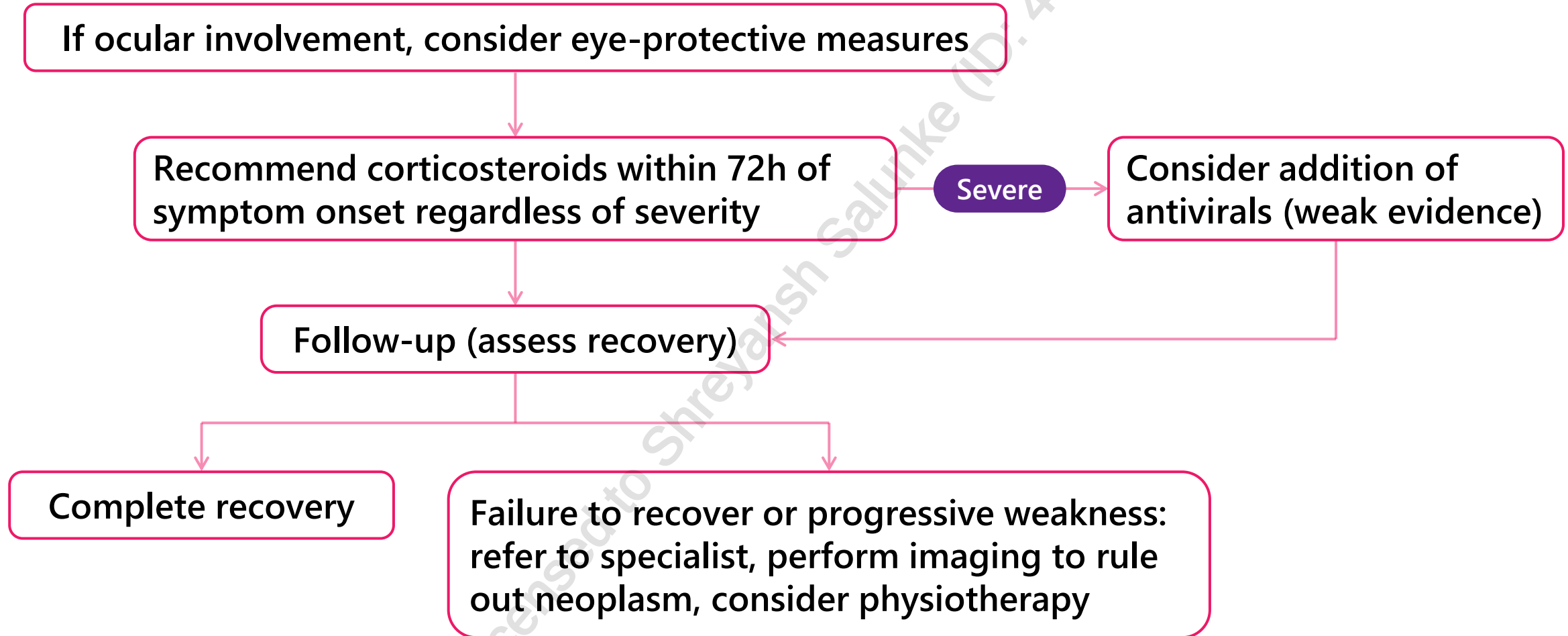
- Reduce the risk of incomplete facial recovery and synkinesis (involuntary face contractions)
- Early treatment (within 72 hours of symptom onset) reduces the risk of developing denervation and can help reduce pain
- One trial demonstrated maximal benefit if corticosteroids are administered within 48 hours of symptom onset
- Steroid choice and dose varies, below are some options
- **No benefit in children**

Drug	Dose	Side Effects
Prednisone	60 mg/day for 5 days, then tapered over another 5 days (for a total dose of at least 450 mg)	Hyperglycemia, GI upset, mood swings, agitation, hypocalcemia, hypokalemia, sodium and fluid retention
Prednisolone	60 mg/day for 5 days, then reduce dose by 10 mg each day for 5 days	



Poor evidence on benefits of adding antiviral therapy in severe cases

BELL PALSY TREATMENT ALGORITHM FOR ADULTS



SPECIAL POPULATIONS

Pregnancy

- Bell Palsy is more common in pregnant women than non-pregnant women
- Usually presents in the third trimester of pregnancy or within the first few weeks postpartum
- Risk factors include: obesity, chronic hypertension and severe pre-eclampsia
- Non-drug measures and acetaminophen are recommended since most will recover spontaneously
- Corticosteroid use is determined on an individual basis
- Exposure to corticosteroids in the first trimester has a slight increased risk of cleft palate but is deemed likely safe for use within and beyond the second trimester

Breastfeeding

- Corticosteroid use while breastfeeding is deemed safe
- Among the antivirals, valacyclovir and acyclovir are the preferred agents

MONITORING PARAMETERS

Efficacy Endpoints		
Parameter	Degree of Change	Timeframe
Facial expression / nerve function	Full Recovery	Gradual, dependent on severity of condition/cause

Safety Endpoints		
Parameter	Degree of Change	Timeframe
Side effects from drug therapies (steroids: agitation, psychosis, hyperglycemia, etc.)	Prevent	Duration of therapy
Recurrent symptoms	Prevent	Ongoing

GUILLAIN-BARRE SYNDROME

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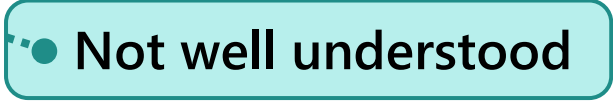
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WHAT IS GUILLAIN-BARRE SYNDROME (GBS)?

- Autoimmune disease in which the body's immune system attacks the peripheral nervous system
- May be triggered by an infection, causing weakness and paralysis of the limbs
- Incidence:
 - Greater in males than in females
 - 1-2 per 100,000 people / year worldwide
 - Affects all age groups although the incidence of GBS increases with each 10-year increase in age past the first decade

PATHOPHYSIOLOGY

- The infection can trigger an immune response that may change the nature or structure of the nervous system
- The immune system will start treating its own cells as foreign and attack the myelin sheath that surrounds the axons of many peripheral nerves
- This damages the peripheral nerves and slows transmission of nerve signals



• Not well understood

CLINICAL PRESENTATION

- Clinical presentation varies for different patients
- Sudden onset and rapid disease progression
- Initial symptoms: bilateral weakness, burning or tingling sensations in legs that can spread to arms, upper body and face
- Delayed symptoms: above symptoms increase in intensity until muscles cannot be used at all, resulting in paralysis (life-threatening)
- Severe symptoms lead to:
 - Breathing problems which may require ventilator support
 - Difficulty swallowing, speaking or chewing
 - Irregular heart rate and blood pressure
 - Issues with bladder or bowel control
 - Unsteadiness when walking

RISK FACTORS

Infections (viral and bacterial)

- Infections involving *Campylobacter jejuni* and *Mycoplasma pneumoniae*
- Cytomegalovirus
- Hepatitis E virus
- Epstein-Barr virus
- Zika virus – GBS is likely triggered by Zika in a small proportion of infections

Vaccinations (influenza or childhood vaccination)

- Rare

Surgery or malignancy

Male gender and older age

GOALS OF THERAPY

- ✓ Reduce the severity of symptoms
- ✓ Accelerate recovery
- ✓ Prevent or treat complications

TREATMENT

- There is no cure for Guillain-Barre syndrome
- Supportive care is very important
 - Many patients are initially admitted to ICU for close monitoring of respiratory, cardiac and hemodynamic function
 - Patients can develop respiratory failure and require mechanical ventilation
 - DVT prophylaxis (LMWH), bladder and bowel care, physical/occupational therapy and psychological support are necessary

DISEASE-MODIFYING TREATMENTS

Plasma exchange (plasmapheresis)

- Process of removing blood from the body, separating RBC and WBC from plasma, returning blood cells back to the patient without plasma, and allowing the body to replace plasma quickly
- Removes antibodies and other immune cells that can contribute to nerve damage
- 4-6 treatments over 8-10 days, each session takes 2 hours
- Demonstrated efficacy when started within 4 weeks of onset
- Side effects: hypotension and sepsis

High-dose immunoglobulin therapy (preferred treatment)

- Intravenous immunoglobulins neutralize antibodies and dilute them to reduce the immune attack on the nervous system
- Demonstrated efficacy when started within 2 weeks of onset, procedure done for 5 days
- Side effects: aseptic meningitis, rash, acute renal failure



Both options equally effective in patients presenting within 2-4 weeks of neuropathic symptoms, no benefit to combining options, either may be used in pregnancy

LONG-TERM OUTCOMES FROM TREATMENT

- 80-84% of patients with GBS can walk independently at 6 months – 1 year after diagnosis
- Relapse is possible within 2 months-10 years after initial episode
- Full recovery of motor strengths occur in 70% of patients after a year
- Small proportions of patients will die despite intensive care
- Patients sometimes need psychological counselling to help them adapt

IMMUNIZATION

- Rarely, GBS cases may occur after administration of tetanus toxoid-containing vaccines
 - Studies have not supported any causal association
- Patients who develop GBS within 6 weeks of vaccination should not receive any further doses
- Immunizations should be withheld for one year after the onset of GBS
- No causal association between GBS and seasonal influenza vaccination
 - In 1976 there was a small increased risk of GBS following an influenza vaccine that was made to protect against a swine flu virus
 - **Absolute risk of GBS is about 1 case / 1 million seasonal flu vaccinations today (very small risk)**

CASE PRACTICE

AJ is a 34-year-old female who is 32 weeks pregnant. She presents to the emergency department with weakness in her face. She reports that the weakness started suddenly this morning on the right side of her face. She noticed difficulty brushing her teeth, difficulty closing her right eye and drooling from the right corner of her mouth. She also had a tingling sensation in her fingers since the previous day, which she assumed was due to her pregnancy. The physical exam confirms no changes in vision or weakness in the limbs. She has no medical conditions. Her blood pressure is 112/70 mmHg, on average, and she does not smoke or drink alcohol. She is anxious about the effect this may have on her baby, and mentions she had a mild cold sore last week.

1. Which clinical finding in this patient makes GBS less likely?
 - a) Recent cold sores
 - b) Tingling sensation in the fingers
 - c) Unilateral presentation of facial symptoms
 - d) Third trimester pregnancy

CASE PRACTICE

2. What is the best initial treatment for this patient?

- A) Oral corticosteroids
- B) Intravenous immunoglobulin (IVIG)
- C) Observation only
- D) High-dose antivirals alone

3. Which non-pharmacological advice can you give the patient to continue at home when discharged?

- A) Facial massage exercises
- B) Using an eye patch and lubricating eye drops
- C) Applying warm compresses to the face daily
- D) Limit fluid intake to avoid edema

4. Which of the following complications is most important to monitor for this patient?

- A) Hearing loss
- B) Corneal ulceration
- C) Seizures
- D) Respiratory failure

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CHANGE LOG

July 2025

- Slide 3: Updated legend
- Slide 10: Slight change to dose of prednisone
- Slide 17: Removed a bullet point due to redundancy
- Slide 22: Added a point
- Slide 24-25: Case added

November 2025

- Minor formatting/rephrasing edits, no content changes made

April 2026

- Slide 2: Updated NAPRA competencies
- Slide 9: Minor update to non-pharmacological measures